Medical Devices That Go Beyond Fitness Trackers And Smart Watches

Demand for medical devices is growing at a fast rate as people become more and more health conscious with increasing government-aided social healthcare schemes and an increasing number of patients moving from hospitals to their homes for nursing and rehabilitation

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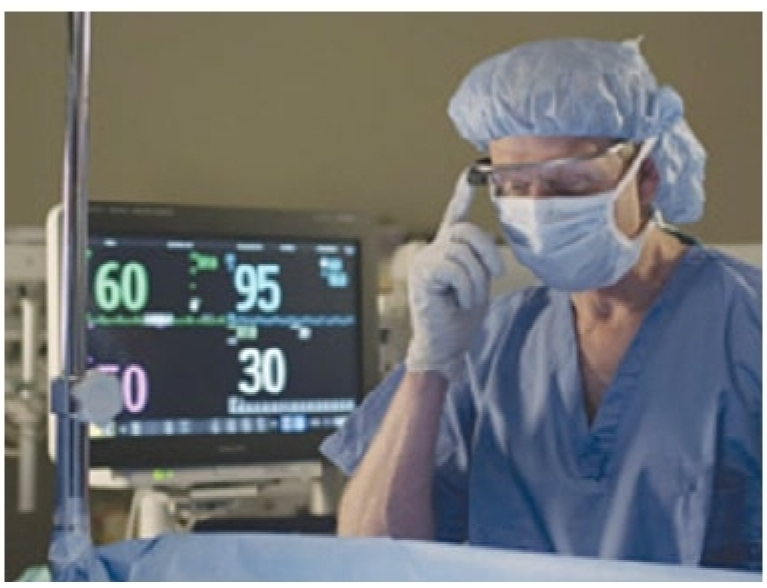
An autonomous, non-invasive medical device that performs a specific medical function such as monitoring or providing support over a prolonged period of time is called a wearable medical device. The term wearable implies that the support environment is either the human body or a piece of clothing. Handheld and portable devices are, strictly speaking, not wearable devices. However, in a broader sense, such medical items as splints, bandages, eyeglasses and contact lenses are commonly considered as wearable medical devices.

Usually, the term wearable medical device simply stands for a sophisticated device with the capability for integrating mechanical functions with microelectronics, computing capabilities and, in some cases, a degree of artificial intelligence (AI). Such a device has the capability to collect, store, process and analyse data to provide necessary feedback, and instantly send alerts or medical decisions to patients and medical supervisors.



An anaesthesiologist keeps his attention on the patient while viewing vital signs via Google Glass (Credit: www.medical.philips.com)

An example of a modern wearable medical device is sleep apnea monitor used for infants susceptible to sudden infant death syndrome. The device consists of a respiratory effort belt worn by the sleeping infant, which provides an alert when breathing patterns become a cause for concern.

Another example is of a navigation device, which is based on global positioning system (GPS), and is used to assist the visually-impaired in navigation, way-finding and obstacle-avoidance.

Wearable devices are designed for specific purposes, such as monitoring vital medical parameters, providing long-term support, medical aid and rehabilitation assistance to patients with temporary or permanent disabilities, non-invasive sensing, communicating with patients and also for receiving feedback from them. Medical parameters being monitored include electrocardiography (ECG), respiration, skin temperature, pulse, blood pressure or blood oxygen saturation, body kinematics, as well as sensorial, emotional and cognitive reactivity for providing effective treatment of cardiac diseases, asthma and diabetes.

Background

The focus in personal healthcare has shifted from hospital care to home care and further to healthcare on-the-move. As a result, usage and popularity are rapidly shifting from fixed medical devices and systems to portable ones and ultimately to wearable devices.

First-generation devices needed user involvement to obtain and record data, which was processed offline; for example, a heart holter, which is used for monitoring and recording cardiac activity. Electrodes placed on the patient's chest are connected to a small battery-operated recording monitor worn by the patient. The patient records the activities undertaken during the period